

Synchronized Robot Movement to Music

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Abstract - Dance is an art form of expression to a specific composition or song. It consists of a sequence of movements either improvised to the music or carefully selected and choreographed to go along with a specific song or part of a song. The beauty in the artform is that it does not just consist of expression, but it has mathematical foundations. To make a dance look pleasing to the audience it must have symmetry, use geometric shapes, and timing. To be able to make a robot “dance” and move synchronised to music, it must contain the balance of symmetry, geometry, and timing. This paper will discuss the difficulties of synchronizing robots and robotic movements to music.

I. Introduction

It is an undeniable fact that robots are growing more and more popular in today’s world. The need and usage for them has increased in the past few years, as has the need for variety and flexibility of movements for robots. Mimicking human ligaments and human movements in a realistic way is a current focus and

problem in robotics today. With a range of different types of human movement, music and dance is the particular topic I chose to focus on.

Music and dance has long been a form of human expression and art. Humans have been working on robots to mimic different arts such as painting and piano playing, but dancing is a difficult problem since it requires the synchronized movement between different ligaments along with the music. Improvised and freestyle dancing is even harder to have a robot perform in the artistic way we want as well. The popular dance move known as “The Robot” acknowledges this fact and it consists of humans moving in a constrained way to mimic the movements of a robot.

By analyzing human movements and the mathematical fundamentals and basis of dancing, we can learn to better robot movement to create a dancing robot in synchronization with music.

II. The Mathematics of Dance

Dance comes in many different forms and styles, but mathematics is a fundamental and basis in dance and

music. Timing, geometric shapes, and symmetry are super important when it comes to beauty and synchronization between a person and dance.

Counting/timing is the most basic and rudimentary starting point for dance and music. One must count to start the music or count to know when to start the dance. The movements of dance are generally divided into 8 counts. If the timing is off between the dancer and the music, the movements do not make sense and will not give off the pretty synchronization between both. This is also the first problem when it comes to synchronizing robot movement to music.

When you start the music, some dances start immediately and some start after a particular count or lyric. When conducting the *Dancing Automata* lab, the most difficult part was not choreographing the movements, but timing the start of the song with the start of the robot movements. We tried to use an LED sequence in our robot to help with the timing of the song, but the problem was starting both the sequence and the music at the same time (Figure 1). At the time of the experiment, possible lags in either the robot or the music were not accounted for.

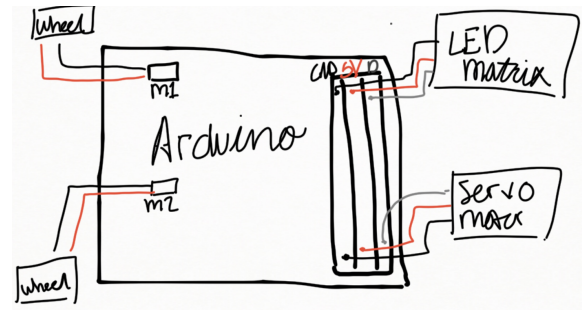


Figure 1: Circuitry of *Dancing Automata* experiment. This circuitry connected LED and servo to robot in order to time the choreography correctly.

A possible solution for this would be focusing on the development of music recognition for the robot. The robot would then listen to the start of the song and when a particular cue happens, then it can begin its dance sequence. Research done by Naoto Nakahara [4] suggests that real-time beat tracking can solve this problem. Although Nakahara's paper focuses on real-time tracking, the same concepts can be applied to a choreographed time. Figure 2 is a diagram that shows the potential solution to solve this counting and robots beat problem.

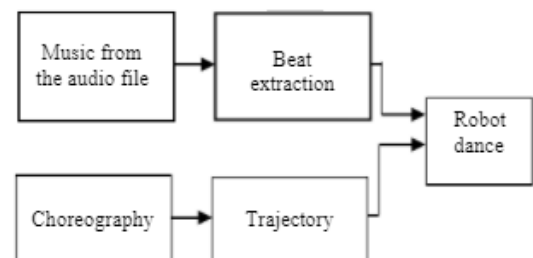


Figure 2: Diagram for synchronous movement

Geometry plays a big role in the beauty of dance. Dancers create straight lines and different shapes such as triangles while performing choreography. [1] Already this is a difficulty when applying the same movements and geometric shapes to robots since we have not achieved the same fluidity of motion in robots. Straight lines are the easiest geometric shapes that robots can be programmed to do in dance. Unlike humans, once the robot is programmed to make straight lines such as put its arms straight in the air, it won't have variations. It would do the same motion in the exact place every single time. Although a lot of shapes that dancers create with their body cannot be reproduced by a robot in a fluid state.

Symmetry is important in dance to make it visually pleasing to the eye. Whether it is the synchronization of people or the symmetry of the individual body, it is important to dance. When synchronizing robots with music, this symmetry is also important not only in the dancing robots, but also in the robot design as well.

In the following sequence of photos, you will see what made the Boston Dynamics robot dancing video so popular and so successful. [6] You can see the synchronization and the symmetry between the robots and the music allowing it to be visually pleasing to the eye. The robots themselves

are also symmetrical in design allowing further symmetry in motion and dance motions.



Figure 3: The starting of the Boston Dynamics robot



The robots in a line using their heads to create circles. It is symmetrical due to the fact that two robots are doing the circle on the left and two are on the right.



The robots showing symmetry with two on the left and two on the right and one in the middle. This composure is commonly found in group dances.

III. Labanotation

The biggest difference between robots and humans when it comes to the arts is feeling. You cannot teach the robot feeling of dance or the feeling of music since it is a programmed being. Rudolf Laban, a Hungarian dancer was the creator of Labanotation. It is a notation system for analyzing human movement. [3] By putting human movement and dance into a notation and mathematical form, it becomes easier to program robots to create the same movements. Labanotation is a way to turn human feelings and movements into mathematical notations in a language that the robots can understand.

Labanotation is measured by three things: the part of the body doing the movement, the duration

of the movement, and the dynamic quality of the movement (Figure 4). By focusing on each individual notion it is attainable to create a robot that can dance in synchronization with music. For example, if focusing on the arm movement of a robot in dance, there are a lot of things to consider such as the rotation of the limb, the duration of the rotation, the speed and other dynamics of the rotation.

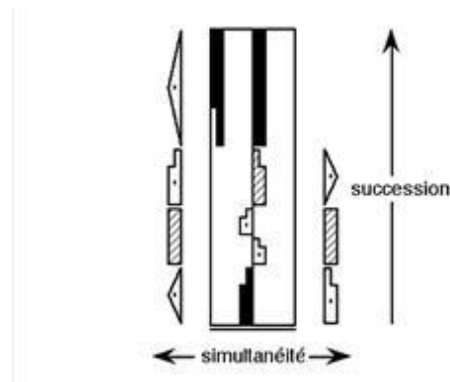


Figure 4: Labanotation notation system

IV. Real World Application

Although a dancing robot is something that many people are looking forward to producing and seeing, the solutions to the various problems in making a dancing robot can further progress research in the field of technology for the disabled. Producing ligaments that move more fluidly, producing robotic arms and legs for the disabled can be made more realistically. The same technology can be applied to other arts and music, not just dance. A pianist with a

robotic hand can play more fluently instead of rigid movements.

The same technology can be applied to therapeutic robots and toys for children allowing them to create learning tools and calming tools for those who need them.

V. Further Work

Researching real-time beat tracking and further continuing other research on having the timing of robots to music will be the main focus on other work. In order to make fully automated or autonomous dancing robots, there are several individual steps that must be done including: music and robotic movement timing, robotic design for optimal fluidity in movement, and labanotion of human movement translated into robotic movement.

References

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